MASTER PROJECT

Shock propagation through the lens of remittances

Who bears the cost? Evidence from US–Mexico migration corridors

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Abstract

This paper studies how localized shocks in migrant destination areas propagate to origin communities through remittance networks. We use Hurricane Ian, which struck Florida in 2022, as a spatially concentrated negative shock to remittance senders in the United States. Combining municipality-level remittance data from Banco de México with migration-network measures based on Mexican consular records, we examine whether Mexican municipalities more exposed to Florida experienced differential changes in remittance inflows after the shock. We then use the structure of migrant networks to study whether exposure to other major U.S. destinations, especially Texas and California, mitigated or amplified the shock. The results speak to how international remittance networks transmit local economic disruptions across borders and to the role of migrant-network diversification in shaping the distribution of these effects.

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Contents

1. Introduction

International remittances are one of the main private channels through which local economic shocks can cross national borders. Mexico is a central setting in which to study this mechanism. Banco de México records US$64.7 billion in family remittance receipts in 2024, while World Bank estimates place Mexico among the largest remittance recipients in the world, second only to India (Banco de México, 2026b; World Bank, 2024). The U.S.–Mexico corridor is especially important because Mexican households receive transfers from migrants whose earnings are tied to local labor markets across the United States. This paper asks whether a negative local shock in the United States is transmitted to Mexican municipalities through remittances. Do migrants smooth transfers after being hit by a shock, or do origin households bear part of the adjustment through lower remittance inflows?

The answer matters because remittances are not only an aggregate balance-of-payments flow, they are also part of household resources. Our data do not directly observe Mexican household income or consumption. Even so, changes in remittance inflows are economically meaningful for household welfare. Prior work shows that remittances are an important source of household resources in Mexico, affecting consumption, poverty, and inequality (Taylor et al., 2005), food security (Mora-Rivera & van Gameren, 2021), and schooling and child labor decisions (Alcaraz et al., 2012). We therefore interpret changes in municipal remittance inflows as reduced-form evidence on a channel with direct relevance for household income and consumption, while not claiming to estimate those household outcomes themselves.

The main empirical difficulty is that most large shocks are not cleanly assigned to one side of the migration corridor. Caballero et al. (2023) show that Mexican households connected to U.S. labor markets more severely affected by the Great Recession were less likely to receive remittances, providing important evidence that migrant networks transmit U.S. shocks to Mexico. Yet the Great Recession simultaneously affected the United States and Mexico, making it difficult to isolate the remittance channel from other aggregate forces. Their analysis also focuses on the extensive margin of remittance receipt. Quantifying how much bilateral remittance flows adjust on the intensive margin is necessary to measure the income loss experienced by origin households.

We address these challenges by exploiting Hurricane Ian, which made landfall in southwestern

Florida on September 28, 2022. Ian caused an estimated US$109.5 billion in damages in Florida and was the costliest hurricane in the state's history (Bucci et al., 2026). The shock was severe and spatially concentrated on the sender side of the corridor: it disrupted economic activity in Florida but did not directly affect Mexican municipalities. Natural disasters in the United States can reduce local productivity, labor demand, housing wealth, and household resources (Boustan et al., 2020). Hurricane Ian therefore provides a localized negative shock to the capacity of Florida-based migrants to remit, while leaving recipient municipalities in Mexico exposed only through their pre-existing migration links to Florida.

We base our argument on the following idea. As a response to a shock to Florida, migrants can directly reduce remittances from Florida to Mexican municipalities. However, many origin municipalities receive remittances from migrants in several U.S. states. If migrants care about total resources available to the origin household, then a fall in Florida remittances can raise the marginal value of transfers from migrants living elsewhere. The same shock can therefore generate two effects at once: a direct decline in Florida-to-Mexico remittances and a compensatory response from other U.S. states connected to the same municipalities. An alternative behaviour of the migrants in the US could also be to domestically remit to the affected state households instead of counteracting the shock through remittances to Mexico. This response assumes that other state migrants are connected to the Florida mexicans. This two reactions may be mutually exclusive, pushing the change in remittances recieved by Mexican municipalities onto opposite directions. The net effect on remittance inflows to Mexican municipalities is therefore an empirical question this paper addresses.

The first part of the paper formalizes this idea with a structural remittance model. The model follows the insight in Albert and Monras (2022) that migrants allocate resources across where they live and where their origin households are located. In our setting, this idea is applied to remittance choices rather than to location choices. Migrants choose between local consumption in the United States and resources available to the origin household in Mexico. They differ in income and in the strength of their altruistic or obligation motive, and remitting involves both a fixed cost and a variable cost. In the spirit of models with heterogeneous agents, fixed costs, and variable costs (Melitz, 2003), the fixed cost generates an extensive margin of remitting, while the variable cost governs the intensity of transfers.

The key extension is that origin-household resources include local non-remittance income plus remittances received from migrants in other U.S. states. This makes the model directly useful for interpreting the empirical patterns. A negative shock to Florida migrants reduces their optimal remittances. At the same time, for households linked to Florida, the fall in Florida transfers lowers perceived origin resources for migrants in other states, raising the marginal utility of additional remittances under an insurance motive. The model therefore provides a unified way to interpret both the direct exposure effect and the compensatory network response observed in the data.

The second part of the paper brings the model to the data by constructing the inflow of remittances flows from U.S. states to Mexican municipalities. Because such flows are not directly observed, we combine two sources from Banco de México with MCAS administrative records. Banco de México reports remittance inflows by U.S. state of origin and by Mexican municipality of receipt (Banco de México, 2026a, 2026c). MCAS records identify the Mexican municipality of origin and U.S. state of residence of Mexican consular-card holders, providing a proxy for the geography of migrant networks. We use these records to construct state-to-municipality migration shares and allocate observed state-level remittance outflows across Mexican municipalities. We focus our analysis on the Mexican municipalities receiving remittances from Florida.

Two validation exercises support this construction. First, the migration network data are highly persistent: the distribution of Mexican migrants across U.S. states and Mexican municipalities has year-to-year correlations above 0.90 from 2012 onward. Second, when we use these migration shares to allocate state-level remittance outflows to Mexican municipalities, the resulting municipal aggregates correlate between 0.75 and 0.85 with official municipal remittance receipts. These facts suggest that the estimated corridors capture meaningful variation in the geography of remittances and that the assumptions underlying our construction are reasonable.

The third part of the paper studies the shock. We exploit variation in Mexican municipalities' pre-existing exposure to Florida to estimate whether Ian affected municipal remittance inflows differently depending on their exposure to the shocked state. In the period immediately following Ian, a 1% increase in a municipality's pre-existing exposure to Florida is associated with a 0.16% decrease in remittances received. This result is consistent with a sender-side income shock being transmitted through the remittance corridor to origin municipalities.

Finally, we examine migration-network spillovers. Among municipalities jointly exposed to Florida and other major sender states, remittances originating from California remain largely unchanged following the shock. By contrast, remittances from Texas, the second-largest sender state, decline in municipalities that are jointly exposed to both migration corridors in the post-Ian period. We explore this relationship further by centering the data on different quantiles of Texas exposure to Mexican municipalities. The diagnostic plots show for the 10% most exposed Florida municipalities to Texas, (insert write-up here), whereas for the least 10% exposed show (describe what we see). This pattern is consistent with both compensatory transfers from migrants in other states and out-migration from Florida-exposed counties.

Related Literature. The paper is close to work that studies remittances as a form of household insurance (Amuedo-Dorantes & Pozo, 2006; Bettin et al., 2025; Yang & Choi, 2007). Most existing evidence studies shocks in receiving countries, where migrants abroad may increase transfers to insure households at origin. We instead study a sender-side shock, where the capacity to remit changes in the destination labor market while origin municipalities are not directly affected. This distinction is important because the same remittance channel that insures households against origin shocks may transmit destination shocks back to those households.

The paper also relates to research on migrant networks and the international transmission of local shocks (Caballero et al., 2023; Munshi, 2003). The closest paper is Caballero et al. (2023), which shows that Mexican households linked to U.S. labor markets hit by the Great Recession became less likely to receive remittances. We differ by studying a localized natural disaster, constructing bilateral state-to-municipality remittance corridors, and focusing on intensive-margin flows and cross-corridor responses.

The structural framework connects to research that models migrants as allocating resources across origin and destination locations (Albert & Monras, 2022). The difference is that our model uses this logic to study remittance behavior after a destination-side shock. It also connects to work on interference and spillovers in causal inference (Aronow & Samii, 2017; Butts, 2023; Hudgens & Halloran, 2008). In our setting, interference is not only a threat to identification, but part of the economic mechanism, because migrants linked to the same origin municipality may jointly insure households against shocks that affect only one destination.

The remainder of the paper is organized as follows. Chapter 2 develops the theoretical remittance model. Chapter 3 introduces Hurricane Ian as the natural experiment and describes both the construction and validation of the migration and remittance data. Chapter 4 presents the empirical strategy and reports the main estimates for exposure effects. Chapter 5 investigates potential spillover effects. Chapter 6 concludes.

2. Theoretical framework and empirical strategy

This chapter outlines the theoretical framework used for our analysis. We begin by adapting a heterogeneous-firm trade model to the context of individual remittance behavior, enstablishing the theoretical foundations of how a migrant's transfer decisions can be influenced by economic shocks.

2.1. A Remittance Model with Network Spillovers and Heterogeneous Migrants

To analyze the impact of shocks on remittance flows, we adapt the logic of heterogeneous-firm trade models to the context of remittances. Specifically, we consider a model where migrants differ in income and altruism intensity, and where remitting requires paying both a fixed access cost and a variable cost proportional to the amount remitted. We then augment the model to allow migrants connected to the same origin municipality to respond to each other's transfers. Formally, consider migrant i, who resides in U.S. state s, originates from Mexican municipality m, and chooses remittances in period t.

We decompose origin-household resources into local non-remittance income and remittances received from migrants in other U.S. states. Perceived origin-household resources excluding migrant i's own remittance are

$$y^{O}_{imst} = \tilde{y}^{O}_{mt} + \sum_{s' \neq s} R_{s'mt}, \tag{2.1}$$

where $\tilde{y}^{O}_{mt}$ denotes local non-remittance income in municipality m, and $\sum_{s' \neq s} R_{s'mt}$ captures remittance inflows from migrants residing in U.S. states other than s. The migrant faces a utility maximization problem under a CES aggregator over local consumption and total resources available to the origin household:

$$U_{imst} = \left[\alpha \left(c^L_{imst} \right)^{\frac{\sigma-1}{\sigma}} + (1-\alpha) \left(\phi_{im} \left[R_{imst} + \tilde{y}^O_{mt} + \sum_{s' \neq s} R_{s'mt} \right] \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}. \tag{2.2}$$

Here $\phi_{im} > 0$ denotes the altruism or obligation intensity parameter, analogous to the productivity parameter in Melitz (2003). This strictly positive parameter ensures that sending remittances can increase the migrant's utility by raising origin-household resources. The variable $c^L_{imst} \geq 0$ represents the migrant's consumption in the destination country, and $R_{imst} \geq 0$ is the amount sent in remittances by migrant i. The weight of local consumption is given by $\alpha \in (0, 1)$, and the elasticity of substitution between local consumption and origin-household resources is given by σ. Origin-household total resources are therefore defined as $R_{imst} + y^O_{imst}$, which we treat as a single composite good. The presence of y^O_{imst} introduces a role for both origin-country economic conditions and migration-network transfers: lower perceived origin resources increase the marginal utility of remittances, generating an insurance motive consistent with Yang and Choi (2007).

The migrant's budget constraint is given by:

$$c^L_{imst} + f_{mt} \mathbb{1}\{R_{imst} > 0\} + (1 + \tau_{mt}) R_{imst} \leq w_{imst}, \tag{2.3}$$

where $w_{imst} \geq 0$ is the migrant's wage, $f_{mt} \geq 0$ is a fixed remitting cost capturing the pre-transaction expense of accessing a remittance channel, analogous to the fixed export cost in Melitz (2003), and $\tau_{mt} > -1$ is the variable cost which scales with the amount remitted, analogous to the iceberg cost structure in Melitz (2003). Notice that, even though τ_{mt} is typically positive to reflect transaction fees or exchange rate spreads, we allow for the possibility of negative variable costs, which could arise from the presence of remittance subsidies or favorable exchange rate regimes.

Assuming that the migrant maximizes utility subject to the budget constraint and chooses to remit a positive amount ($R_{imst} > 0$), solving the Lagrangian yields the optimal remittance function:

$$R^*_{imst} = \frac{A_{imst} \left[w_{imst} - f_{mt} + (1 + \tau_{mt}) \left(\tilde{y}^O_{mt} + \sum_{s' \neq s} R_{s'mt} \right) \right]}{1 + (1 + \tau_{mt}) A_{imst}} - \left(\tilde{y}^O_{mt} + \sum_{s' \neq s} R_{s'mt} \right), \tag{2.4}$$

where we define the parameter A_{imst} as:

$$A_{imst} \equiv \left(\frac{(1-\alpha)\phi_{im}}{\alpha(1+\tau_{mt})} \right)^{\sigma}. \tag{2.5}$$

The term A_{imst} represents the optimal ratio of origin-household resources to own local consumption, adjusted for the variable cost of remitting. It comprises the migrant's altruism intensity (ϕ_{im}), the utility weight of own consumption (α), the elasticity of substitution (σ), and the variable cost of remitting (τ_{mt}).

To understand the effect of destination-country economic shocks on remittance flows, we analyze changes in the migrant's wage (w_{imst}). The partial derivative of optimal remittances with respect to wage is strictly positive:

$$\frac{\partial R^*_{imst}}{\partial w_{imst}} = \frac{A_{imst}}{1 + (1+\tau_{mt})A_{imst}} > 0. \tag{2.6}$$

Therefore, through an income effect, the model predicts that a positive shock to the migrant's wage will increase remittances, while a negative shock will decrease them. The fixed cost f_{mt} establishes an entry threshold: if the migrant's wage is not sufficiently high relative to the fixed cost, they will not send remittances at all. Additionally, the variable cost τ_{mt} reduces the marginal benefit of each dollar sent, dampening the sensitivity of remittances to wage fluctuations.

The augmented model also delivers a spillover prediction. Holding the migrant's own wage and remitting costs fixed, remittances from other migrants enter the decision through perceived origin-household resources:

$$\frac{\partial R^*_{imst}}{\partial R_{s'mt}} = -\frac{1}{1 + (1+\tau_{mt})A_{imst}} < 0 \qquad \text{for } s' \neq s. \tag{2.7}$$

A fall in remittances from one destination lowers perceived resources at origin and raises the optimal remittance from migrants in other destinations. For municipalities highly exposed to the shocked destination, this lowers origin-household resources and increases the marginal utility of additional transfers from migrants residing elsewhere. The model therefore provides a theoretical basis for compensatory migration-network spillovers: the same shock can generate a direct fall in remittances from the affected destination and an offsetting response from other connected destinations.

The model delivers three empirical implications that guide the analysis. First, because optimal remittances are increasing in migrant income, a negative wage or income shock in the country where migrants reside should reduce remittances sent by affected migrants. Second, origin municipalities with greater pre-existing exposure to the affected migrant destination should experience larger declines in total remittance receipts, all else equal. This exposure effect is a reduced-form object: it captures the direct loss in transfers from the affected destination net of any adjustment by migrants in other destinations. Third, because remittances from other destinations enter the migrant's decision through perceived origin-household resources, a decline in transfers from the affected destination raises the marginal value of remittances from migrants in alternative destinations connected to the same municipality. The model therefore predicts compensatory responses from other high-sending destinations, especially for municipalities jointly exposed to the affected destination and those alternatives. These implications map directly into the empirical strategy below: we first estimate whether remittances fall from the shocked destination, then test whether municipal receipts fall more in places more exposed to that destination, and finally examine whether alternative corridors increase transfers relative to the shocked corridor among jointly exposed municipalities.

3. Empirical Context and Data

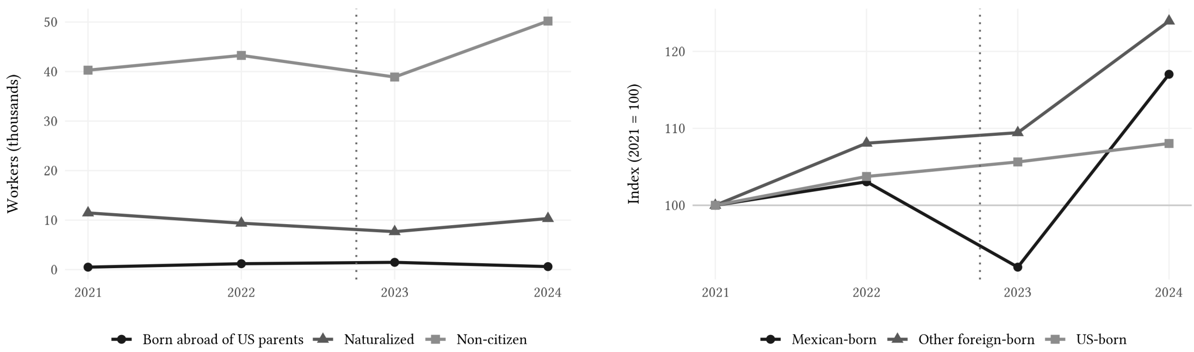
This chapter introduces the empirical context and data used in our analysis. We begin by introducing our source of exogenous variation in remittance flows, Hurricane Ian. After documenting the hurricane's impact on Florida's remittance outflows, we explain the data sources and the methodology we use to construct our treatment measure, which is used to differentiate Mexican municipalities by how exposed they are to the shock. Finally, we present a validation exercise, justifying the choice of our constructed treatment variable as a proxy for the Mexican migrant population in the United States.

3.1. The Natural Experiment: Hurricane Ian (2022)

Hurricane Ian struck the southwestern coast of Florida on September 28, 2022, with peak sustained winds of 140 kt (around 260 kph) (Bucci et al., 2026). Ian caused an estimated US$ 112.9 billion in total damage in the United States, of which US$ 109.5 billion occurred in Florida alone, making it simultaneously the costliest hurricane in Florida's history and the third-costliest in US history. In Lee County alone, at least 52,514 structures were impacted, of which 5,369 were completely destroyed (Bucci et al., 2026).

Beyond its aggregate toll, Ian's destruction fell on sectors that employ a large share of Mexican migrants. Construction is the single largest sector of employment for Mexican-born workers in Florida, at roughly 28% of all those employed, and the four sectors most exposed to the storm together account for over 50% of Mexican workers. Within them, construction makes up 52%, agriculture 17%, landscaping 16%, and hospitality 15%.[1] Construction is therefore both the main source of work for Mexican workers and the channel where Ian's damage translates most directly into lost employment, and later into reconstruction demand.

Figure 3.1b indexes Florida construction employment to 2021 by workers' region of birth. The Mexican-born series diverges from the US-born and other foreign-born series after Ian, falling to roughly 92% of its 2021 level in 2023 before rebounding to about 117% in 2024. The US-born series drifts upward throughout, which suggests the disruption was specific to migrant labor rather than to construction as a whole. Figure 3.1a shows that this movement is carried by non-citizens, who make up the large majority of Mexican-born construction workers in the state and absorb both the 2023 dip and the 2024 rebound. Non-citizens are also the group most reliant on remittances as a financial link to their origin households (Handlin et al., 2002). The descriptive evidence thus indicates that Ian affected workers whose earnings finance remittances to Mexico, which motivates the migrant income channel we examine in the remainder of the paper.



(a) Mexican-born construction workers by citizenship status

(b) Construction employment by region of birth, indexed to 2021

Figure 3.1: Mexican-born construction employment in Florida around Hurricane Ian. Source: IPUMS USA, ACS 1-year samples 2021–2024; employed Florida construction workers. Dotted line marks Hurricane Ian (2022Q4).

[1] Calculated using IPUMS USA, ACS 1-year samples 2021–2024, employed Mexican-born residents of Florida. The full industry composition and the within-sector shares are reported in the online appendix.

Crucially for our identification strategy, Hurricane Ian did not affect Mexico. The storm originated in the Caribbean, made landfall in Cuba and Florida, and dissipated over the Atlantic after a final less impactful landfall in South Carolina (Bucci et al., 2026). Therefore, the shock is spatially clean. Indeed, remittance senders in Florida were directly exposed to a severe and sudden wealth shock (reference or data?), while remittance receivers in Mexican municipalities were entirely unaffected. This allows us to better isolate the transmission of the shock through the remittance channel. In the language of the model, Hurricane Ian is the negative shock to migrants' economic resources in the country where they reside, Florida is the affected destination, and Mexican municipalities differ in their pre-existing exposure to Florida through migration networks.

To motivate the analysis, Figure 3.2 presents the evolution of remittance outflows from Florida to Mexico using data from Banxico. A sharp decline in remittance outflows can be observed in the quarter immediately following Hurricane Ian, with flows falling by approximately 40% relative to the previous quarter. This pronounced contraction suggests that the hurricane generated a substantial disruption in remittance activity from Florida to Mexico, thereby providing a strong motivation for examining its effects more formally.[2]

[Figure pending]

Figure 3.2: Remittance outflows from Florida to Mexico, in millions of US dollars. The vertical dashed line indicates the quarter of Hurricane Ian (Q4 2022).

Source: Authors' own elaboration based on Banxico SIE data.

3.2. Data Sources

Given that bilateral remittance flows are between US States and Mexican municipalities are not publicly available, we base our estimation strategy on multiple data sources. The source of

[2] Florida outflows are plotted against the average outflows of all other U.S. states. Because the comparison series is constructed as an average, potential opposite shocks affecting individual states during the Hurricane Ian period may be attenuated. Nevertheless, the purpose of the figure is simply to illustrate the magnitude of the decline observed in Florida.

remittance data is Banco de México (Banxico), through the datasets provided by its Sistema de Información Económica (SIE). Specifically, series CE166 records remittance inflows received by each Mexican municipality. This series is available at quarterly frequency, from Q1 2013 to Q1 2026 and represents our outcome variable (Banco de México, 2026c). To capture the Mexican migration network, we utilize administrative records from the Matricula Consular de Alta Seguridad (MCAS) program, managed by the Instituto de los Mexicanos en el Exterior (IME) (Instituto de los Mexicanos en el Exterior, 2026). The publicly available version of this dataset provides a comprehensive registry of consular identification cards issued to Mexican nationals residing in the United States each year, from 2010 to 2024. While the MCAS does not constitute an exhaustive census of all migrants, it represents a robust proxy for the spatial distribution of Mexican migrants in the United States. In particular, it is highly representative of undocumented and recently arrived migrants, who face the strongest incentives to obtain consular identification (Caballero et al., 2018).

3.3. Migration Networks Construction

To build our treatment measure, we construct a migration-based matrix using annual MCAS from 2010 to 2024. Because these records capture new consular issuances rather than the complete stock of the remitting population, which also includes previously arrived migrants, we rely on the assumption of persistance of migration patterns over time. To mitigate the volatility of yearly migration fluctuation and construct a robust proxy for the long-term spatial distribution of Mexican migrants, we compute the average of these matrices across all years.

Formally, we define $MCAS_{ms}$ as the average annual count of MCAS issuances to migrants from Mexican municipality m residing in U.S. state s. For each mexican municipality, we calculate the average migration network share, w_{ms}, which represents the proportion of municipality m's total US migrant population that resides in state s. This is calculated as the ratio of $MCAS_{ms}$ to the total number of MCAS issuances for migrants from municipality m across all U.S. states S: Mexican population originating from the m municipality, as the ratio of $MCAS_{ms}$ to the total number of MCAS issuances from that state across all municipalities M:

$$w_{ms} = \frac{MCAS_{ms}}{\sum_{s=1}^{S} MCAS_{ms}} \tag{3.1}$$

By construction, the sum of w_{ms} across all U.S. states for a given Mexican municipality m

equals 1, and the value of w_{ms} for a specific state s reflects the relative importance of that state as a destination for migrants from municipality m, which in turn is a proxy for the exposure of municipality m to shocks occurring in state s.

3.4. Persistence Assumption and Validation

The validity of this estimation relies on the assumption of temporal persistence of migration flows, implying that the distribution of migrants recorded in the 2010–2024 MCAS data is representative of the active remitting population. This assumption is critical to our estimation strategy, as it justifies the use of MCAS data to construct the migration-based weighting matrix in order to allocate remittance flows. The stability of migration patterns over time is well-documented in the literature, with studies showing that enstablished migration networks tend to dictate future migration patterns, creating persistent flows between specific origin and destination pairs (Card, 2001; Munshi, 2003).

To verify this assumption within our specific sample period, we conduct a validation exercise using the MCAS data by computing the Pearson correlation of our constructed weighting matrices across years. As shown in Figure 3.3, when taking any year from 2012 onward as the base year, the correlation with subsequent years is consistently above 0.90. While the 2010 and 2011 matrices exhibit strong correlation between them, their correlation with later years is smaller, specifically between 0.75 and 0.85. This may be due to structural adjustment in migration patterns following the Great Recession.

[Figure pending]

Figure 3.3: Correlation of MCAS-based weighting matrices across years. Each point represents the correlation coefficient between the weighting matrix of the base year and the weighting matrix of the comparison year.

Source: Authors' own elaboration based on MCAS data.

This high degree of correlation suggests that the spatial distribution of Mexican migrants across U.S. states has remained highly stable over time, lending credibility to our use of an averaged weighting matrix for estimating bilateral remittance flows. Given that Hurricane Ian hit florida in Q4 2022 and the MCAS data for Florida in 2013 is missing, to construct our treatment variable we use the average of the weighting matrices from at the end we used the covid years to construct the average? @ zucca 2014 to 2021, which are all highly correlated with each other, and avoid any endogeneity in the treatment caused by the shock.

4. Empirical Methodology and Results

4.1. Estimation Strategy

The primary challenge in estimating the impact of the Hurricane Ian shock on remittance inflows to Mexican municipalities lies in the sorting of migrants. Municipalities with higher exposure to Florida may differ systematically from those with networks concentrated in other U.S. states. Therefore, a standard cross-sectional analysis comparing municipalities with different levels of exposure to Florida would likely lead to biased estimates due to confounding factors correlated with both the treatment (i.e. exposure to Florida) and the outcome (i.e. remittance inflows). This classic omitted variable bias problem makes a causal interpretation of cross-sectional estimates impossible (Wooldridge, 2010). To overcome this challenge, we leverage the temporal variation in remittance inflows by employing a continuous Difference-in-Differences (DiD) event-study design (Callaway et al., 2024), where our treatment is the pre-shock exposure to Florida, measured by $w_{m,\text{FL}}$.

4.2. Event-Study Specification

We use the following Poisson Pseudo Maximum Likelihood (PPML) event-study specification to analyze the impact of the Hurricane Ian shock on remittance inflows to Mexican municipalities:

$$\mathbb{E}[R_{mt} \mid \mathbf{X}_{mt}] = \exp\left(\alpha_m + \gamma_{s(m),t} + \sum_{k \neq -1} \beta_k \cdot \left(w_{m,\text{FL}} \cdot \mathbb{1}\{t - T_0 = k\}\right)\right) \tag{4.1}$$

This methodology is motivated by Silva and Tenreyro, 2006, who show that the PPML estimator remains consistent in the presence of heteroskedasticity and excess zeros, which are common features of remittance data at the municipality level. The event-study specification allows us to estimate the dynamic treatment effects of the shock over time. We include municipality fixed effects (α_m) to control for all time-invariant, unobserved municipal characteristics that might otherwise introduce omitted variable bias. Additionally, we incorporate state-by-time fixed effects ($\gamma_{s(m),t}$), where $s(m)$ denotes the specific Mexican state to which municipality m belongs. These fixed effects account for both macroeconomic shocks common to all regions and time-varying unobservables at the state level. The interaction term $w_{m,\text{FL}} \cdot \mathbb{1}\{t - T_0 = k\}$ captures the differential effect of the shock on Mexican municipalities based on their varying degrees of baseline exposure to Florida, where $w_{m,\text{FL}}$ is a measure of pre-shock exposure to Florida and $\mathbb{1}\{t - T_0 = k\}$ is an indicator for each time period relative to the shock. The coefficients β_k thus represent the change in remittance inflows for a one-percent increase in Florida exposure at each time period k relative to the shock. Errors are clustered at the municipality level to account for potential correlation within municipalities over time.

4.3. Results

Figure 4.1 shows the result of the PPML event-study estimation, where the vertical dashed line represents the timing of the shock (Q4 2022). The pre-shock coefficients (β_k for $k < 0$) are close to zero and statistically insignificant, visually consistent with the parallel trends assumption underlying our DiD design. The point estimate in the quarter of the shock (Q4 2022) is negative and statistically significant at the 1% level, indicating that municipalities with higher exposure to Florida experienced a larger drop in remittance inflows immediately following the shock. Specifically, a one percentage point increase in Florida exposure is associated with a 0.16% decrease in remittance inflows in the quarter of the shock. In the subsequent quarters, the coefficients become slightly positive but statistically insignificant at the 5% level, suggesting that the initial negative impact of the shock on remittance inflows dissipated over time. These results suggest that the Hurricane Ian shock had a significant, but short lived impact on remittance flows, suggesting a rapid re-adjustment of remittance networks after the shock (connection to the model?). However, the fact that the point estimates in the post-shock period are positive, altough not statistically significant, may be indicate a potential overshooting of remittance inflows due to post-shock reconstruction period, or some form of spillover effects through migrant networks in other states, which we will investigate in the next chapter.

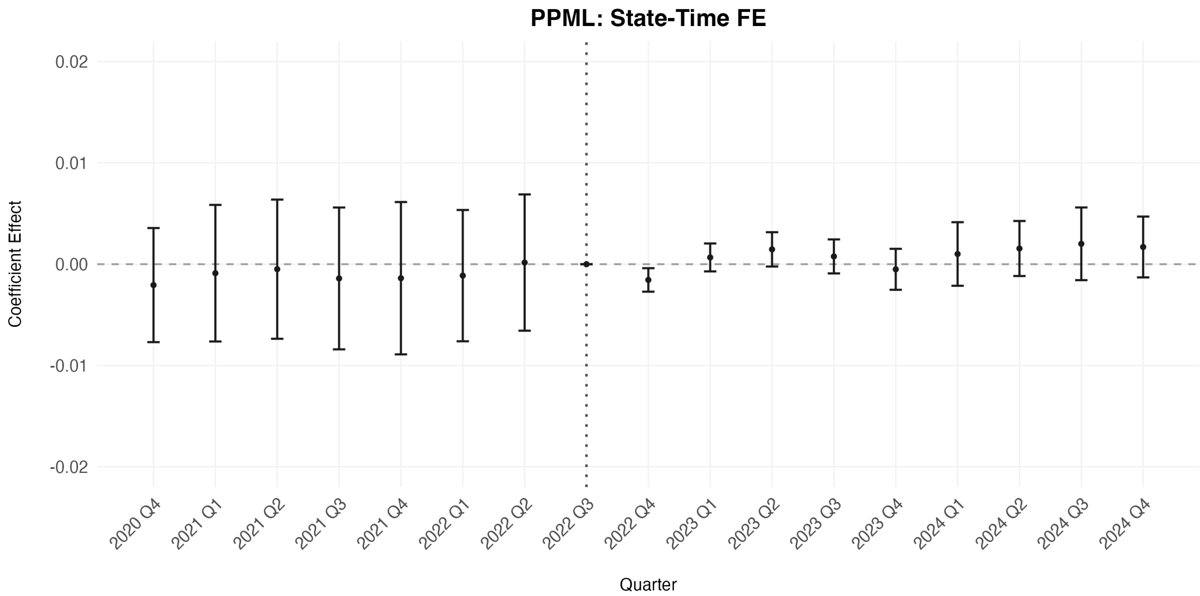


Figure 4.1: PPML Event Study Estimates

As a robustness check, we also estimate the same event-study specification using spatially clustered errors, following the methodology of Conley (1999), to account for potential spatial correlation in remittance inflow shocks affecting neighboring municipalities. The results, shown in Figure 4.2, are qualitatively similar to the baseline specification, with the main difference being the thighter confidence intervals in the preperiod as buffer distance increases, and slightly wider in the post-shock period. Given that the results are robust to both clustering methods, and that the spatial clustering specification doesn't produce positive definite variance-covariance matrices for buffer distances above 50km, we consider the municipality-level clustering as our baseline specification.

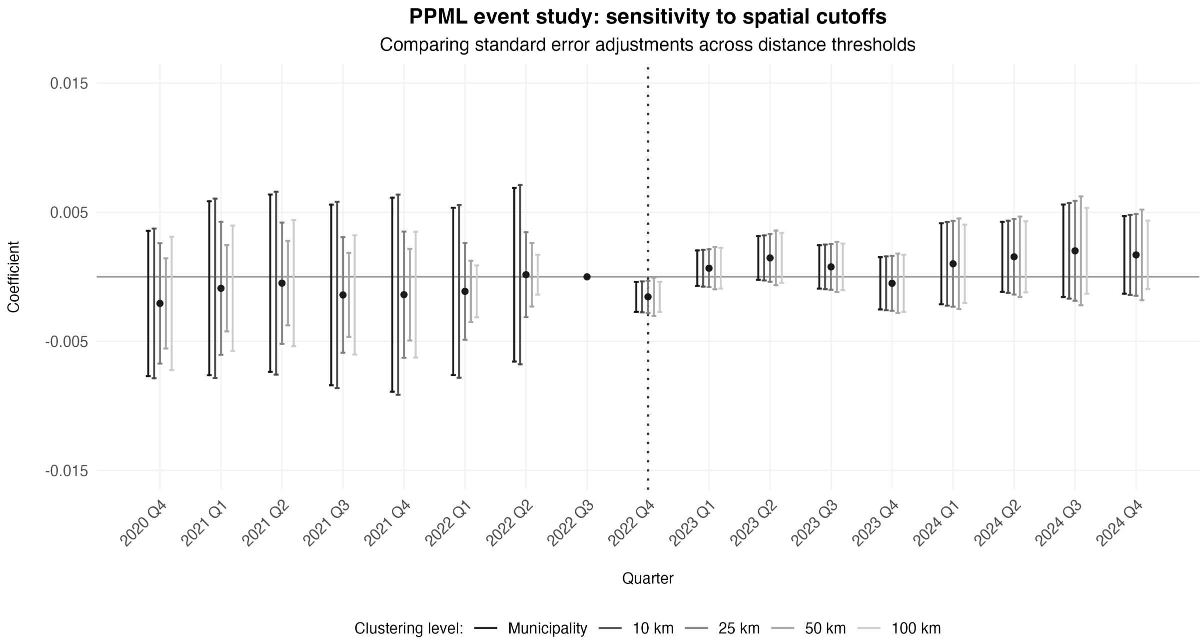


Figure 4.2: PPML Event Study Estimates with Conley Standard Errors for Different Buffer Distances

4.4. Is the Response Specific to Florida?

A natural concern is that $w_{m,\mathrm{FL}}$ could proxy for general migration intensity rather than Florida exposure specifically. If so, the impact-quarter contraction could reflect a broad migration-municipality trend coinciding with 2022Q4 rather than the hurricane. We address this by augmenting the specification with the analogous exposure to the two other largest destination states, California and Texas, each interacted with the full set of event-time indicators. Under the confounding story, a common migration trend would move all three exposure paths together at the shock; under our interpretation, only exposure to the state Ian actually struck should react.

Conditioning on California and Texas exposure leaves the Florida impact coefficient essentially unchanged at -0.0013 ($p = 0.034$), while the California and Texas impact coefficients are themselves flat and insignificant ($+0.0004$ and -0.0001, respectively). The contraction at landfall is thus specific to Florida exposure and is not correlated with migration to other major destinations.

4.5. Robustness Checks

The consistency of the PPML estimator relies only on the assumption that the conditional mean is correctly specified (Silva & Tenreyro, 2006). Unlike the log-linear model, it requires no assumption on the conditional variance, so heteroskedasticity and overdispersion do not threaten consistency; they affect only efficiency and inference, which we handle through standard errors clustered by municipality. This narrows the robustness exercise to two questions: whether the exponential conditional mean is well specified, and what the mean-variance structure of the data looks like.

We address the first with a Ramsey RESET test as proposed by Ramsey (1969) and applied to this setting by Silva and Tenreyro (2006), which augments each specification with the square and cube of its fitted index and tests their joint nullity. A rejection signals that the assumed conditional mean is misspecified. We find that a log-linear OLS specification strongly rejects correct specification at all conventional levels ($W = 835.10, p < 0.001$), whereas the PPML specification does not reject the null ($W = 1.18, p = 0.308$). The RESET evidence therefore favors the exponential conditional mean over an additive-in-logs alternative.

The second question is answered by Figure 4.3, which plots, for each municipality, the variance of its pre-shock quarterly remittance inflows against their mean, both in logs. On these axes the Poisson benchmark is the line of slope one, along which the variance rises one-to-one with the mean. The cloud lies well above this line and follows a slope of approximately 1.546, so the variance scales faster than the mean, signaling overdispersion. This leaves PPML unaffected, since its consistency does not depend on the variance function. A log-linear model fares worse: the expected value of log inflows is not a linear function of the regressors, so estimating the model in logs by OLS does not recover the parameters of the conditional mean (Silva & Tenreyro, 2006). The two diagnostics thereby support PPML as the baseline estimator.

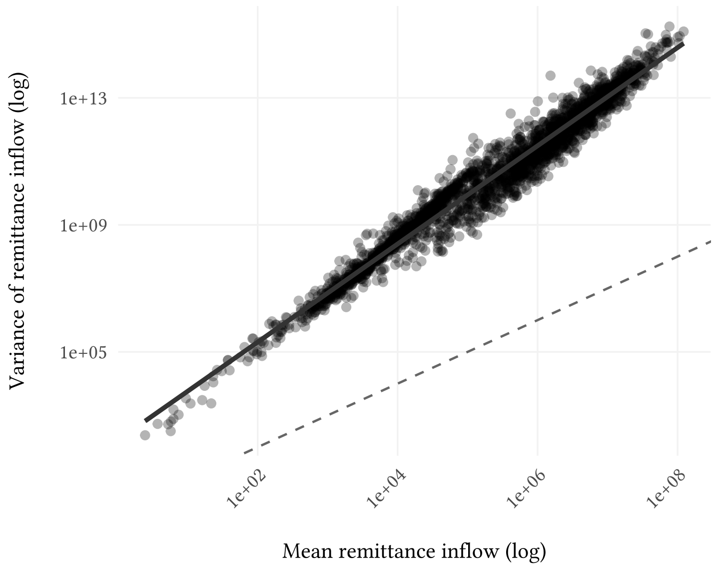


Figure 4.3: Mean-variance relationship of municipality remittance inflows. Each point represents a Mexican municipality; axes show the variance against the mean of its quarterly inflows over the pre-shock window. The solid line is the fitted relationship; the dashed line marks the Poisson equidispersion benchmark, where variance equals the mean.

4.6. Sensitivity Analysis

The pre-shock coefficients in Figure 4.1 are small and statistically insignificant, which is consistent with parallel trends. However, tests for pre-trends are known to have low power, so a failure to detect a pre-trend is necessary but not sufficient to conclude that none is present; conditioning estimation and inference on having passed such a test can itself distort the results (Roth, 2022). We therefore choose to not rest identification on the flat pre-period alone. Rather, we ask how large a violation of parallel trends the impact-quarter estimate could withstand.

In doing so, we follow the relative-magnitudes approach of Rambachan and Roth (2023). Rather than imposing exactly parallel trends, it bounds the post-shock violation by $\bar{M}$ times the largest deviation observed in the pre-shock window, and traces the robust confidence set as $\bar{M}$ grows. The breakdown value is the largest $\bar{M}$ for which that set still excludes zero, measuring how large a post-shock violation the significant effect can absorb before becoming indistinguishable from zero. A value $\bar{M} = 1$ represents a post-shock violation as large as the worst pre-shock deviation, which serves as a conventional lower bound for robust results.

For the baseline Florida-only specification, the impact-quarter effect breaks down at $\bar{M} = 0.15$. The effect is thus distinguishable from zero only if any post-shock differential trend is at most about fifteen percent as large as the largest pre-shock deviation, far short of the conventional value of one. The netted specification of Section 4.4, which isolates Florida from correlated migration, is necessarily less precise (its significance falls to $p = 0.034$), and a less precisely estimated effect reaches zero under a smaller assumed violation, breaking down at $\bar{M} = 0.04$. Across spatial-clustering choices the breakdown remains in a comparable range, with Conley cutoffs of 10, 25 and 50 km reported in the online appendix.

We therefore read the evidence cautiously. The sign and timing of the response, a sharp Florida-specific contraction at landfall that is absent for the placebo exposures to Texas and California, are credible and consistent with remittances transmitting the income shock through the region where migrants earn. The magnitude is fragile, since small departures from parallel trends suffice to overturn its significance.

5. Investigating Spillover Effects

A fundamental requirement for standard causal inference models is the Stable Unit Treatment Value Assumption (SUTVA), formalized by Rubin (1980). SUTVA assumes that the potential outcomes of any unit depend exclusively on its own treatment assignment, excluding the possibility of any indirect spillover between units.

In our context, U.S. states not directly exposed to a local shock might still be affected if control units are exposed to the treatment through indirect channels, leading to biased estimates (Clarke, 2017). For instance, given that California and Texas all host a big proportion of Mexican migrants, we hypothesize that these two states might have played a role in offsetting the negative Florida shock. If a Mexican municipality highly dependent on Florida remittances experiences an income shock due to Hurricane Ian, connected migrants residing in Texas might respond by increasing their transfers to compensate for the drop. If this was the case, the estimated Florida-exposure effect on total remittances would capture the net effect of the shock after within-network adjustment, rather than the direct fall in Florida-origin remittances.

Therefore, we argue that our estimates are subject to general equilibrium effects. We base our analysis on the exposure-mapping logic of Aronow and Samii (2017) to define indirect exposure through alternative migrant networks. Rather than assuming that a municipality's remittances respond only to Florida exposure, we allow the response to vary with exposure to other major U.S. destinations, especially Texas and California.

5.1. Empirical Strategy

The biggest hosts of Mexican migrants are Texas and California, representing the 20% and 30% respectively, of all Mexican migrants yearly arriving to the US; thus, it is likely that those municipalities exposed to the Florida (FL) shock might also be connected to a certain extent to California (CA) and Texas (TX). We test the possible compensation from the states of TX and CA using a triple interaction strategy, using the PPML specification as already deployed in previous sections:

$$\begin{aligned}\mathbb{E}[R_{mt}|X] = \exp\Big(&\alpha_m + \gamma_{st} + \beta_1(\text{Exposure}_m^{FL} \times \text{Post}_t) + \beta_2(\text{Exposure}_m^{TX} \times \text{Post}_t) \\ &+ \beta_3(\text{Exposure}_m^{FL} \times \text{Exposure}_m^{TX} \times \text{Post}_t) + \epsilon_{mt}\Big)\end{aligned} \tag{5.1}$$

As constructed above, the measure of exposure $Exposure_m^{FL}$ represents the percentage of migrants from the origin municipality m that have moved to FL. Analogous measures can also be constructed for the states of TX and CA. The equation above accounts for municipality fixed effects (α_m), Mexican state-quarter fixed effects (γ_{st}) and clustered standard errors at the municipality-quarter level.

In this setting,

$$\left.\frac{\partial \log \mathbb{E}[R_{mt} \mid X]}{\partial Exposure_m^{FL}}\right|_{Post=1} = \beta_1 + \beta_3 Exposure_m^{TX}$$

Meaning that β_1 captures the post-shock effect of FL exposure when TX exposure is 0, while β_3 captures how the effect of FL exposure changes with TX exposure. In other words, β_3 allows us to explore how a continuous change in TX exposure affects the response of FL remittances after the FL shock. If $\beta_1 < 0$ and $\beta_3 > 0$, this would suggest that the negative effect of FL exposure on remittances is mitigated for municipalities with higher TX exposure, indicating a compensatory effect from the TX migrant network.

5.2. Results

The results of the regression are shown in Figure 5.1 and the coefficients of interest β_3 are reported in Table 5.1. From Figure 5.1 we can see how for both CA and TX, the pre-trends are flat before the Hurricane event, suggesting parallel trends that would validate our identification strategy. When it comes to the coefficients reported in Table 5.1, we notice different patterns across the two states. In the CA case, we observe that the panel stays essentially flat around zero, which suggests that no significant spillover effects were found. As for TX, we see a slightly negative, significant and persistent decline of the coefficient, meaning that the more exposed a given municipality is to TX, the more pronounced is the negative post-shock effect of the FL exposure on remittance patterns. In other words, TX exposure is associated to a larger net decline in remittances after the FL shock. MAYBE CONSIDER RESCALING THE FIGURE SO WE SET UPPER AND LOWER BOUNDS OF +- 0.02 so the negative effect is more appreciated?

[Figure pending]

Figure 5.1: Evolution of the coefficient β_3 over time for the interaction between FL and TX (left) and FL and CA (right). The vertical dashed line indicates the timing of the Hurricane Ian shock. The shaded area represents the 95% confidence interval.

Since this result a priori does not support the compensatory spillover hypothesis of the model, we next aim to explore how the post-shock FL effect on remittances varies according to different levels of TX exposure. That is, 5.2 plots the effect

$$\beta_1 + \beta_3 \text{Exposure}_m^{TX}$$

by inserting different values of $Exposure_m^{TX}$ (specifically, the 10th and 90th percentiles of the distribution of TX exposure across municipalities). The figure shows that the net effect of the FL shock on remittances is more negative for municipalities with higher levels of TX exposure, which is consistent with the negative and significant β_3 coefficient found above. This reinforces the idea that instead of a compensatory effect, we observe an amplification effect, where the negative impact of the FL shock on remittances is more pronounced for municipalities that are also highly exposed to TX.

Table 5.1: Network Spillovers: Texas vs. California

Dependent Variable:	remittances	
	Texas Network	California Network
Model:	(1)	(2)
Variables		
fl_tx_interaction × period_date = 2020-10-01	7.44×10^{-5}	
	(0.0001)	
fl_tx_interaction × period_date = 2021-01-01	-3.24×10^{-6}	
	(0.0002)	
fl_tx_interaction × period_date = 2021-04-01	2.67×10^{-5}	
	(0.0002)	
fl_tx_interaction × period_date = 2021-07-01	0.0001	
	(0.0001)	
fl_tx_interaction × period_date = 2021-10-01	0.0001	
	(0.0002)	
fl_tx_interaction × period_date = 2022-01-01	4.38×10^{-5}	
	(0.0001)	
fl_tx_interaction × period_date = 2022-04-01	-2.69×10^{-5}	
	(0.0001)	
fl_tx_interaction × period_date = 2022-10-01	6.09×10^{-5}	
	(5.46×10^{-5})	
fl_tx_interaction × period_date = 2023-01-01	-0.0002***	
	(7.11×10^{-5})	
fl_tx_interaction × period_date = 2023-04-01	-0.0002***	
	(7.01×10^{-5})	
fl_tx_interaction × period_date = 2023-07-01	-0.0002**	
	(7.06×10^{-5})	
fl_tx_interaction × period_date = 2023-10-01	-0.0001	
	(0.0001)	
fl_tx_interaction × period_date = 2024-01-01	-0.0004**	
	(0.0001)	
fl_tx_interaction × period_date = 2024-04-01	-0.0004**	
	(0.0001)	
fl_tx_interaction × period_date = 2024-07-01	-0.0005***	
	(0.0001)	
fl_tx_interaction × period_date = 2024-10-01	-0.0005***	
	(0.0001)	
fl_ca_interaction × period_date = 2020-10-01		-3.75×10^{-5}
		(0.0002)
fl_ca_interaction × period_date = 2021-01-01		0.0002
		(0.0002)
fl_ca_interaction × period_date = 2021-04-01		0.0001
		(0.0001)
fl_ca_interaction × period_date = 2021-07-01		0.0002
		(0.0001)
fl_ca_interaction × period_date = 2021-10-01		0.0002
		(0.0002)
fl_ca_interaction × period_date = 2022-01-01		0.0002
		(0.0001)
fl_ca_interaction × period_date = 2022-04-01		0.0001
		(0.0001)

[Figure pending]

Figure 5.2: Net Florida shock evaluated at low and high levels of Texas exposure. The figure compares the implied post-shock Florida effect for municipalities at the 10th and 90th percentiles of Texas exposure.

We believe this pattern could be explained by internal transfers or internal mobility across the US states. That is, it could be the case that the Mexican migrants residing in TX try to help their family/friends residing in FL after the shock. Thus, the money that would have been sent back to Mexico is now being sent internally to FL, explaining why we see a decline in remittances received in municipalities with high exposure to both FL and TX. A similar argument could be made for the internal mobility cause: migrants residing in FL could decide to move away after the shock and relocate to TX, which is close and also a major host of Mexican migrants. This would lead to a decline in remittances received by municipalities with high exposure to both FL and TX, since we believe that relocated households have (at least temporarily) less capacity to send money back to Mexico. These two concerns are not accounted in the theoretical model.

Our hypothesis formulated above are consistent with past literature examining domestic transfers and migration in the context of disasters. Although most of this literature so far is focused on developing countries, we believe that the network mechanisms to support affected relatives and smooth consumption are likely to operate similarly in developed contexts. For instance, Savage and Harvey, 2007 and Mozumder et al., 2009 provide important evidence regarding internal transfers as a mechanism of informal insurance arrangement and examine the behavioural response of unaffected regions towards disaster-affected ones. Existing studies in developed countries have also investigated the perspective on internal migration as a response to shocks. Fussell et al., 2023 use migration data around Hurricanes Katrina and Rita, both catastrophic natural disasters affecting US states along the Gulf of Mexico in 2005. Out-migration from the affected counties increased roughly four percentage points over unaffected coastal counties. The response was not a one-year spike but rather persistent, which the authors interpret as residents reducing their exposure to repeated hazard risk by leaving the coast. A storm of comparable scale therefore makes net out-migration from Ian-affected Florida a reasonable prior. The highest-exposure Florida residents have the strongest incentive to relocate to lower-risk states, and the Katrina and Rita evidence shows both an immediate displacement and a multi-year elevation in out-migration from the hardest-hit areas.

6. Conclusion

6.1. Summary of Findings

The main objective of this study was to investigate the impact of a local shock, specifically Hurricane Ian, on remittance patterns from U.S. states to Mexican municipalities. Our analysis revealed a significant negative effect of Florida exposure on remittances, indicating that municipalities with higher exposure to Florida experienced a more pronounced decline in remittance inflows following the hurricane. In fact, the estimated effect suggests that a 1% increase in Florida exposure from the Mexican municipality is associated with a 0.16% decrease in remittances, which is a substantial impact given the importance of remittances for many Mexican households. This findings are consistent with the theoretical approach provided in the model, which predicts that a negative shock in a major U.S. destination can lead to a decline in remittances to the origin municipality, especially for those with higher exposure to the affected state.

To investigate the potential spillover effects from other major U.S. destinations, coherently with what the model would suggest, we explored the role of Texas and California as alternative migrant networks that could potentially compensate for the negative shock in Florida. However, our analysis did not find evidence of a compensatory effect from California, while for Texas we observed an amplification effect, where the negative impact of the Florida shock on remittances was more pronounced for municipalities with higher exposure to Texas. This suggests that instead of mitigating the negative shock from Florida, Texas exposure may have exacerbated it. More concretely, the estimated spillover effect from Texas indicates that a significant 1 percentage-point increase in Texas exposure amplifies the negative post-shock effect of Florida exposure by approximately 0.04%.

To theoretically explain this results the model should account for not only the mexican families but also the migrant networks in the U.S. hitted by a shock, to be potential transfer recievers.

The altruition logic in the model would also apply in the shock setting, treating those mexican U.S households as needy remittances recievers. The amplification effect observed would be consistent under the assumption that .... TO BE COMPLETED

Finally, our findings suggest that Caballero et al., 2023 results on the persistence of remittance patterns after a shock might not be generalizable to all contexts, specially when we analyse a localized event. Their measured impacts may combine destination-country income shocks with origin-country demand effects generated by the global nature of the Great Recession, meanwhile, the shock used by this paper is clean and localized, allowing us to isolate the direct effect of the shock on remittance patterns without confounding factors.

6.2. Policy Implications

The findings of this study have important policy implications for both Mexican and U.S. policymakers. For Mexican policymakers, the significant decline in remittances following the Florida shock highlights the vulnerability of Mexican households that rely heavily on remittance inflows although a relevant finding is also the non persistance of the Florida exposed remittances decline. This suggests a need for policies that can provide inmediate alternative sources of income or financial support to affected households, especially in municipalities with high exposure to U.S. destinations but without the necessity to give a long term support. For U.S. policymakers, the results underscore the importance of considering the broader economic impacts of local shocks, such as natural disasters, on migrant communities and their families back home. Policies aimed at providing support to affected migrants could help mitigate the negative spillover effects on remittance flows by reducing the internal transfers between connected migrants across states.

6.3. Limitations and Future Research

While this study provides valuable insights into the impact of local shocks on remittance patterns, it also has several limitations that should be acknowledged. First, our analysis is based on a single shock event (Hurricane Ian), which may limit the generalizability of our findings to other types of shocks or different contexts. Future research could explore the effects of various shocks, such as economic downturns or policy changes, on remittance patterns to provide a more comprehensive understanding of the dynamics at play. Second, our measure of exposure is based on the percentage of migrants from each municipality residing in specific U.S. states, which may not fully capture the complexity of migrant networks and their influence on remittance behavior. Future studies could incorporate more detailed data on migrant networks, including social ties and communication patterns, to better understand how these factors mediate the impact of shocks on remittances. Finally, while we explored potential spillover effects from Texas and California, there may be other states or factors that contribute to the observed patterns. Future research could investigate additional channels through which shocks in one state may affect remittance flows from other states, such as labor market conditions or policy responses in those states.

I would also include: the fact that remittance sent vary depending on how many immigrants go in and out of the states, which is data we do not have (and that could refine our analysis). Also, the fact that we do not have data on internal transfers between migrants in the US.

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